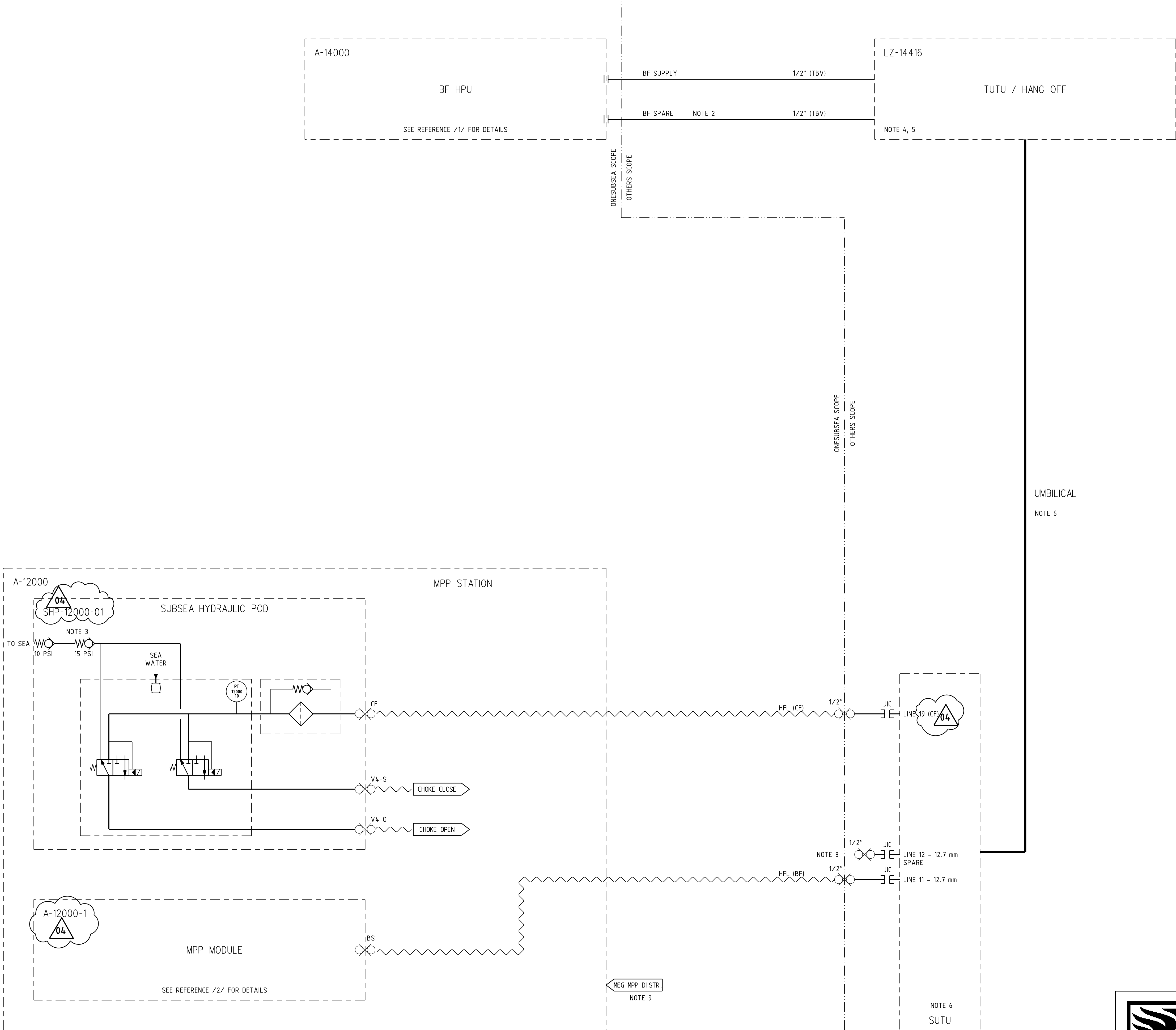




Document Title:			MPP SYSTEM – System Engineering – Hydraulic Distribution Schematic													
Package Description:			SUPPLY AND INSTALLATION OF BOOSTER PUMP PACKAGE													
Equipment Tag Number(s) or Stock Code Number(s):			EA-A-12000													
Asset:			EIDER ALPHA													
TAQA Bratani Ltd. Job Number:			AEIA0280													
Project Title:			EIDER/OTTER MULTIPHASE PUMP													
Suppliers Name:			ONESUBSEA PROCESSING AS													
Suppliers Order Number:			P-GTQ02 - 9412													
Suppliers Document No:	Suppliers Revision:	Total No. of Pages (Incl. Front Sheet):	Date:	Suppliers Comments:												
DRW-0005975	F	2	02-Jan-2018	Reissued for Acceptance												
TAQA Bratani Ltd. Supplier Document Status: <table border="1"> <tr> <td>1</td> <td>ACCEPTED As Final Certified</td> <td>X</td> </tr> <tr> <td>2</td> <td>ACCEPTED: Work May Proceed, incorporate Comments And Re-Submit</td> <td></td> </tr> <tr> <td>3</td> <td>REJECTED: Work May Not Proceed, Revise And Re-Submit</td> <td></td> </tr> <tr> <td>4</td> <td>ACCEPTED: For Information Only</td> <td></td> </tr> </table> Acceptance does not relieve the Supplier/Sub-Contractor of any contracted responsibility of the correctness or quality of work warranty and guarantee noted or implied in the order.			1	ACCEPTED As Final Certified	X	2	ACCEPTED: Work May Proceed, incorporate Comments And Re-Submit		3	REJECTED: Work May Not Proceed, Revise And Re-Submit		4	ACCEPTED: For Information Only		TAQA Bratani Ltd. Comments: Comments made do not imply acceptance of any cost or delivery impact.	
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4	ACCEPTED: For Information Only															
Department		Print Name		Sign	Date											
Projects		E Wright		<i>Ewan A Wright</i>	17/01/18											
Contract Number	TAQA Bratani Ltd. Purchase Order No.		SDRL Code	Sequence Number	Submission											
77IFS	120058		C21	0003	04											



NOTES

- DELETED
- BARRIER FLUID SPARE LINE TO BE PRESSURE COMPENSATED.
- NON-RETURN VALVE IN POM-MATERIAL TO PREVENT MARINE GROWTH AND SCALING IN RETURN LINE (TBD).
- TUTU SHALL BE EQUIPPED WITH MANOMETER AND SAMPLING POINT (COMPANY SCOPE).
- TOPSIDE VALVE AND TUBING ARRANGEMENT MUST ALLOW FOR "ROUND-TRIP" FLUSHING OF BF LINES IN UMBILICAL PRIOR TO FIRST TIME INSTALLATION.
- LINE SIZES AND NUMBERING AS GIVEN IN 77-AEIA0280-Y-SE-0002-000 APPENDIX C.
- DELETED
- BF SPARE LINE DUMMY CAP IS COMPANY SCOPE
- MEG SUPPLY FROM DOWNLINE. SEE REF /3/ FOR DETAILS

LEGENDS

	HYDRAULIC LINES		INSTRUMENT PACKAGE
	HYDRAULIC JUMPER		SCOPE BOUNDARY
	SECONDARY / PILOT HYDRAULIC LINES		
	ISOLATION VALVE, NORMALLY OPEN		ISOLATION VALVE, NORMALLY CLOSED
	PRESSURE COMPENSATOR		FLANGED CONNECTION
	SPRING LOADED CHECK VALVE		DIRECTIONAL CONTROL VALVE
	JIC INTERFACE		FILTER
	HYDRAULIC STAB CONNECTOR		FIELD INSTRUMENT
	DIVER / ROV OPERATED ACTUATOR		

ABBREVIATIONS

BF	BARRIER FLUID
CF	CONTROL FLUID
HFL	HYDRAULIC FLYING LEAD
MPP	MULTIPHASE PUMP
PT	PRESSURE TRANSMITTER
SUTU	SUBSEA UMBILICAL TERMINATION UNIT
TBD	TO BE DETERMINED
TBV	TO BE VERIFIED
TUTU	TOPSIDE UMBILICAL TERMINATION UNIT
V4-O	V4-OPEN
V4-S	V4-SHUT / CLOSE

REFERENCES

- /1/ 77IFS-120058-C21-0002-000 (DRW-0005956) BF HPU Hydraulic Schematic
/2/ 77IFS-120058-C21-0004-000 (DRW-0006426) Pump Barrier Fluid Schematic
/3/ 77IFS-120058-C01-0001-000 (DRW-0005922) Process Flow Diagram
/4/ 77IFS-120058-C22-0003-000 (DOC-0107979) Hydraulic Systems Description

DESIGN BASIS

DESIGN PRESSURE (BF HPU)	124 bara
DESIGN PRESSURE (SUBSEA)	99.3 barg
DESIGN TEMPERATURE	80/-10 °C
DESIGN TEMPERATURE (PUMP OPERATING)	80/4 °C
DESIGN PRESSURE (CF SYSTEM)	230 barg
MAXIMUM WORKING PRESSURE (CF SYSTEM)	207 barg
MINIMUM OPERATING PRESSURE (CF SYSTEM)	90 barg
CONTROL FLUID:	MacDermid Pelagic 100

REV	DATE	DRAWN BY	ENG BY	CHKD BY	APPR BY	DESCRIPTION
04	02-Jan-2018	TSO	-	AVE	CH	RE-ISSUED FOR ACCEPTANCE
03	01-Nov-2017	TSO	-	AVE	CH	RE-ISSUED FOR ACCEPTANCE
02	24-Aug-2017	PBT	PBT	CH	TSK	RE-ISSUED FOR ACCEPTANCE
01	22-Jun-2017	PBT	-	CH	OAS	ISSUED FOR ACCEPTANCE

C.P. No : TAQA-SC-1497
C.P. Title : Supply and Installation of Subsea Booster Pump Package

TAG No.

EA-A-12000

DRAWING No.

77IFS-120058-C21-0003-000



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Title	MPP SYSTEM - System Engineering Hydraulic Distribution Schematic	Weight	N/A
ID	DRW-0005975	Scale	NTS
Format	A1	Sheet	1 of 1
Revision Status	Published	Made by	Checked by
F Released	2018-01-02 15:17:16 TSO	AVE	CH
Engineering Change Order: NA			
SAP Doc version: NA			